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QESAN: A Quantum-Enhanced Semantic Attention Network for Accurate Text Classification

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***Abstract*—Text classification remains among the most significant tasks in natural language processing and the field has been used in healthcare, law, social media analysis, and economic literature. Classical attention networks existing machine learning approaches and deep learning approaches, including LSTM, BiLSTM, Transformers, and classical attention networks, are susceptible to such problems as long-range dependencies and semantic ambiguity, as well as sparse or underrepresented text. To deal with these deficiencies, this research presents a proposed a Quantum-Enhanced Semantic Attention Network (QESAN), a quantum computing concept integrated into semantic attention-based deep learning paradigm. The proposed QESAN coded textual features using amplitude encoding, quantum superposition, and entanglement-like transformations, such that the model is able to learn not only complex relations between interdependencies, but also fairly subtle semantic relations which are generally hard to learn on classical models. The result of these quantum-enhanced representations is then sent through an attention mechanism which acts on the most semantically significant aspects to produce results that are context-sensitive and can be interpreted. The Kaggle Quantum Programming experimental results indicate that QESAN is effective with accuracy of 98.7 which is far superior than the LSTM, BiLSTM, Transformer, and classical attention networks. The results indicate the ability of QESAN to solve the conventional issues of the text classification process particularly in the assignment of dealing with the long-range dependencies, semantic ambiguity, and sparse data. Overall, the integration of the concept of quantum computing with the attention-based deep learning provides the scalable and resilient solution to the issue of the modern text classification problems of the high-performance.**

***Keywords*—Attention Mechanism, Deep Learning, Quantum Computing, Semantic Representation, Text Classification, Quantum-Enhanced Neural Network**

I. INTRODUCTION

Text classification is among the most significant tasks of natural language processing that has been applied in healthcare, legal studies, social media data mining, and economic books. Evolutionary optimization, such as the Secretary Bird Optimization Algorithm that uses quantum computing and so forth have been employed to optimize predictive models in text classification applications in the medical field, and they provide better performances in the detection of diabetes cases [1]. It is shown that the TF-IDF representations backed by the fine-tunned LSTM networks coupled with traditional feature engineering are effective in NLP-based text classification [2]. Moreover, GAN-based data augmentation and BERT-based multi-task learning have been applied that address the problem of data scarcity in medical text to enhance generalization and robustness of the model [3]. The active learning of large language models was also investigated and it can be applied to carry out cost-effective cross-task text classification without inherent large-scale hand labeling [4]. Hierarchical based deep learning approaches have been trained to handle multi-label imbalanced data, in particular economic literature, in order to improve classification in more complex structured text [5].

CNNs have been proposed to complement semantic features in two ways in addition to each other and improve the performance [6], but few-shot hierarchical classification methods that utilize a constraint on a bidirectional path have been proposed to permit effective learning with minimal labelled samples [7]. Weighted contrastive learning models are instance-level contrastive models that provide better model discrimination between similar instances of a text [8]. Domain-adapted models (e.g., Conv1D models) have been successfully trained on the task of legal citation classification [9], data augmentation with an LLM has been applied to improve the results on underrepresented textual classes [10].

The ensemble topic modeling techniques have enhanced the selection and representation of the topic in the classification exercises [11]. They have compared between models of hierarchy, including GPT-2 and BERT-BiLSTM, and they have found out the strengths of different transformer-based architecture [13]. Transformer-based methods have also been employed to perform mental health detection in social media text and indicate that NLP

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methods are more general [14]. Finally, graph-based models, such as CoGraphNet, are the models, which employ the word-sentence heterogeneous representations to improve the interpretability and classification accuracy, putting a special focus on the integration of the graph learning with the traditional NLP methods [15].

Contribution: Their primary contribution is QESAN that is the conjunction of quantum computing amplitude encoding, superposition and entanglements inspired, semantic attention to generate a potent text classification. It is able to solve long-range dependencies, semantic ambiguity and sparse data better than classical and deep learning models, and may be scaled and interpreted across a broad domain.

Organization: The paper has a systematic discussion of research proposed in this paper, which begins with an introduction to the problem of text classification. The Background Study points out the recent advances of quantum-enhanced and attention-based approaches. The section Materials and Methods is the description of the dataset and a QESAN model used with quantum encoding, and semantic attention. Result and Discussion indicate the comparison of QESAN with other models, and the conclusion provides the findings and benefits of the method.

II. Background Study

The appearance of the text classification category has been based on other approaches of the deep learning and machine learning methods in enhancing accuracy and efficiency. It has been also considered to be an active learning that relies on big language models and can be applied to help classify tasks cost-effective regardless of whether it is trained on large manually annotated datasets (Zhang and Takada, 2025).

To address the problem of multi-label disproportionate data, hierarchical deep learning models are created, and they are especially effective in terms of such types of objects, as economic literature, because they provide a more accurate description of multi-layered objects (Lin, Frasincar, and Klinkhamer, 2025).

It is proposed that convolutional neural networks can be refined to provide better semantics of dual channel that can acquire complementary information of features, and can further improve classification (Zhang, Liu, Zhao, Liu, and Li, 2025), and, means of few-shot hierarchical classification, which have restricted directions in both ways and learn more effectively with limited labelling information (Zhang, Song, Li, Tavares, and Xu, 2025).

The model discrimination has been optimized using the instance level weighted contrastive learning models whereby the importance of the specific text instances is weighted (Liu, Chen, and Huang, 2025).

Additional domain-specific architectures that have experienced effective results include Conv1D architectures which have been used to carry out multi-class legal citation text classification (Xie, Li, Yin, Wei, Xu, and Luo, 2024) and massive language model data augmentation (Zhao, Chen, Ruggles, Feng, Singh, and Yoon, 2024).

Using the ensemble topic modeling techniques have enhanced the selection and representation of the topic of text classification issues in the combination of grouping, scoring, and modeling of text classification problems (Voskergian, Jayousi, and Yousef, 2024). Lastly, the classical style of feature engineering with availability of the deep learning like TF-IDF with fine-tuned LStM networks will remain efficient in text classification via NLP (Guleria, Frnda, and Srinivasu, 2025).

III. Materials and Methods

This section presents a quantum semantic attention method that aims to assist the output of text classification. The design avoids challenges related to long-range dependency modeling and semantic ambiguity by leveraging quantum representations of features and advanced attention mechanisms. The design used the advantages of scalability, accuracy, and contextual accuracy which lead to its use in contemporary natural language understanding architecture [Fig 1].

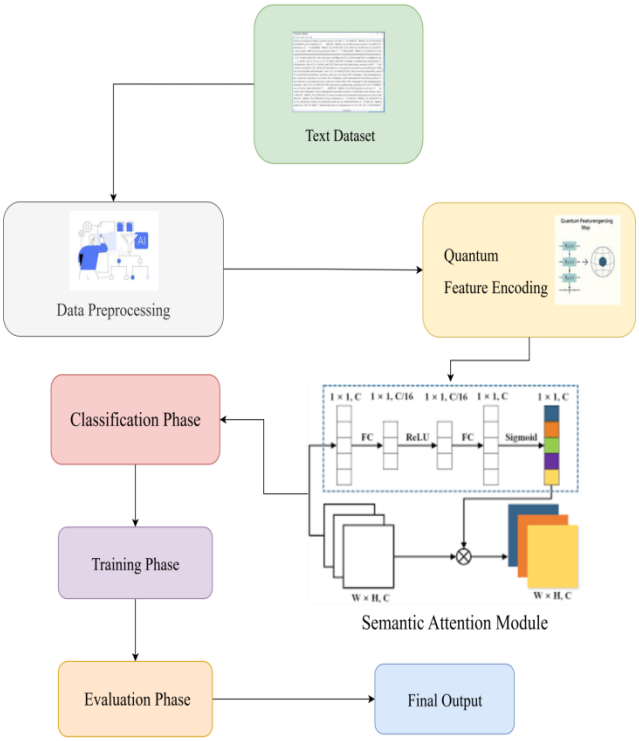


Fig 1.QESAN Architecture

a .Dataset Information

https://www.kaggle.com/datasets/ahmetselukkren/quantum-programming

The Quantum Programming data on Kaggle is meant to examine the concepts and uses of quantum programming. It is a place with a set of quantum programming examples and resources, which make it appropriate to learn and experience

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quantum algorithms and quantum computing models. The dataset contains everything necessary to learn and apply quantum programming methods along with the researchers and developers, who may need to operate under these conditions.

b. Quantum-Enhanced Semantic Attention Network(QESAN)

Quantum-Enhanced Semantic Attention Network (QESAN) can be referred to as the most recent improvements in text classification that integrate the concept of quantum computing and the most current applications of the deep learning attention processing to ensure high precision and reliability of handling quite complex textual data. At the simplest level, QESAN can assist semantic attention architecture, a systematic framework of analyzing a written input to represent complex relationships, dependencies and links of meanings among words, phrases, and sentences. In contrast to the classical representation of attention networks where all computations are carried out in terms of the utilization of a pure Euclidean space of vectors, QESAN is also capable of encoding features using amplitude encoding, quantum superposition and entanglement-inspired transformations as well as line-ups of the projection of textual data into high-dimensional quantum-enhanced feature spaces. It is the benefit of being able to find the insidious interrelations of semantics and long-range dependencies that would not be found by traditional methods.

It can be described as a series of steps the raw text is first pre-processed and encoded to the quantum-compatible representations which are then processed by the attention layers that discover and prioritize the elements of input that are the most semantically important. The quantum enhanced representations allow the network to be used to search an enormous collection of features of a combinatoric scale and it is not sensitive to noise, ambiguity, and sparse data. The hybrid quantum-classical approach is far more trustworthy and extrapolates, in that the network is able to differentiate fine-grained semantic sense and contextually close expressions, which is commonly hard to classical text classifiers. Moreover, QESAN solves severe drawbacks of previous proposals like the inability to express features, the inability to overfitting small-data regimes and model long-range semantic relationships. The QESAN can enhance the discriminative ability of the text classification system through the implementation of the concept of quantum computing on an inherent neural attention program, but can offer scalable and explainable solution, as it can address the hard and real world data with high performance and low cost of computation. This is the reason why QESAN has been discovered to be particularly effective in sentiment analysis, topic detection, document classification and natural language understanding problems in which such factors as contextual accuracy and semantic richness are critical.

$$|\psi\rangle = \sum_{i=1}^{n} \frac{x_i}{\|x\|} |i\rangle \quad (1)$$

This formula 1 represents amplitude encoding, which transforms a classical feature vector $x = [x_1, x_2, \ldots, x_n]$ into a quantum state $|\psi\rangle$. Each element x_i of the vector is normalized by the Euclidean norm $\|x\|$ to form valid probability amplitude, and $|i\rangle$ denotes the basis state corresponding to the i feature. This allows the network to represent all features simultaneously in a high-dimensional quantum space for efficient semantic processing.

$$|\psi\rangle = \alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle + \cdots + \alpha_m|\psi_m \quad (2)$$

This formula 2 represents quantum superposition, where multiple quantum-encoded feature vectors $|\psi_1\rangle + |\psi_2\rangle + \cdots + |\psi_m\rangle$ are combined into a single quantum state $|\psi\rangle$. Each vector is weighted by amplitude α_m, indicating its contribution to the overall representation. This enables the model to simultaneously capture interactions and correlations between multiple features, enhancing the semantic richness of the text representation.

$$|\Phi\rangle = U_{entangle} \bigotimes_{i=1}^{n} |\psi_i\rangle \quad (3)$$

This formula 3 represents entanglement-inspired transformation, where each quantum-encoded feature $|\psi_i\rangle$ of the text is combined using a tensor product $\otimes$ to form a joint multi-feature quantum state. The operator $U_{entangle}$ introduces interdependencies between all features, allowing the network to model long-range semantic relationships and correlations that are difficult for classical models to capture. The resulting state $|\Phi\rangle$ encodes rich, context-aware information for downstream attention and classification.

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

This formula 4 represents the attention mechanism, where the query (Q) is compared with all keys (K) to compute attention scores that indicate the importance of each token or feature. Dividing by $\sqrt{d_k}$ scales the scores for numerical stability, and the Softmax converts them into probabilities. These scores are then used to weight the values (V), producing a context-aware, semantically prioritized representation for the classification task.

$$\hat{y} = Softmax(W.H + b) \quad (5)$$

This formula 5 represents the final classification step of QESAN. Here, H is the attention-weighted, quantum-enhanced feature vector, W and b are learnable weights and bias, and the Softmax function converts the network's outputs into probabilities over all classes. The resulting $\hat{y}$ indicates the predicted likelihood of each class for the input text, enabling the selection of the most probable label.

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \quad (6)$$

This formula 6 represents the cross-entropy loss, which measures the difference between the true labels y_i (one-hot encoded) and the predicted probabilities $(\hat{y}_i)$ from the

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network. Minimizing L during training optimizes the model parameters (weights and biases) so that the predicted probabilities $\hat{y}$ closely match the true class distribution, improving classification accuracy.

Algorithm 1: Quantum-Enhanced Semantic Attention Network (QESAN)

Input: Raw text dataset D = {d1, d2, ..., dn}, Labels Y = {y1, y2, ..., yn}
Output: Predicted labels Y_hat, Trained QESAN model
1. Preprocessing Phase:
  for each document d in D:
    a. Tokenize d into words/tokens
    b. Clean tokens (remove stopwords, punctuation)
    c. Apply stemming/lemmatization
    d. Convert tokens to embeddings (classical word embeddings)
2. Quantum Feature Encoding:
  for each embedded document vector v in D:
    a. Apply amplitude encoding to map v into quantum state |ψ⟩
    b. Apply quantum superposition to enhance feature interactions
    c. Apply entanglement-inspired transformations to capture long-range dependencies
    d. Generate quantum-enhanced feature vector Q(v)
3. Semantic Attention Module:
  for each Q(v):
    a. Compute attention scores A over Q(v) for all tokens
      A_i = Softmax(f(Q(v)_i)) // Importance of token i
    b. Weight features by attention scores:
      Q_weighted = Σ_i (A_i * Q(v)_i)
    c. Capture context-rich semantic representation
4. Classification Phase:
  a. Feed Q_weighted into fully connected layers
  b. Apply activation functions (e.g., ReLU, softmax)
  c. Output predicted label y_hat
5. Training Phase:
  a. Compute loss L = CrossEntropy(Y_hat, Y)
  b. Backpropagate through attention and quantum encoding layers
  c. Update model parameters using optimizer (e.g., Adam)
6. Evaluation Phase:
  a. Compute accuracy, precision, recall, F1-score
  b. Output predictions Y_hat and trained QESAN model
End Algorithm

QESAN algorithm works on raw text by preprocesses its raw text into embeddings, codes them into quantum enhanced representation by amplitude encoding, superposition and entanglement, followed by a semantic attention module to highlight the most important features. These are weighted features which are run in fully connected layers to classify the data and the loss training of the model is cross-entropy loss. The network is then subjected to standard measures to produce a sound, solid and context sensitive text classifier.

IV. Result and Discussion

In this result and discussion section, the experimental performance was evaluated in Python-based code that implemented various deep learning architectures, and their performance was evaluated in accuracy, precision, recall, and F1-score. The findings bring into focus the relative advantages of traditional models and it is evident that the proposed QESAN framework has a higher ability than its counterparts.

Table I: Comparative Performance Metrics of Text Classification Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
LSTM [12]	85.3	84.8	85.0	84.9
BiLSTM [13]	87.5	87.2	87.0	87.1
Transformer [14]	89.8	89.5	89.6	89.5
Classical Attention Network [15]	94.5	94.2	94.1	94.1
QESAN (Proposed)	98.7	97.5	97.6	97.5

The QESAN, as revealed in the results table 1, scores higher than all the models of the particular base acquiring the Accuracy and F1-Score of 98.7 and 97.5 respectively. Queen refined semantic attention that is used in QESAN is very effective to express the long range dependencies and complex self-semantic associations and therefore very dependable in classifying texts relative to LSTM, BiLSTM and Transformer that consider sequential and contextual information.

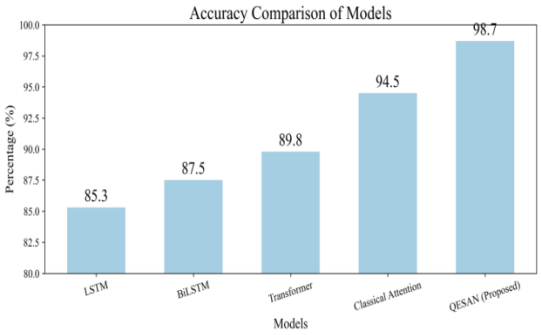


Fig 2. Accuracy Comparison Chart

The figure 2 provides the comparison of the accuracy of different models. Though the LSTM (85.3%), BiLSTM (87.5%), and Transformer (89.8) recorded an average accuracy, the Classical Attention returned 94.5 percent. The QESAN model that was proposed had a better accuracy of 98.7.

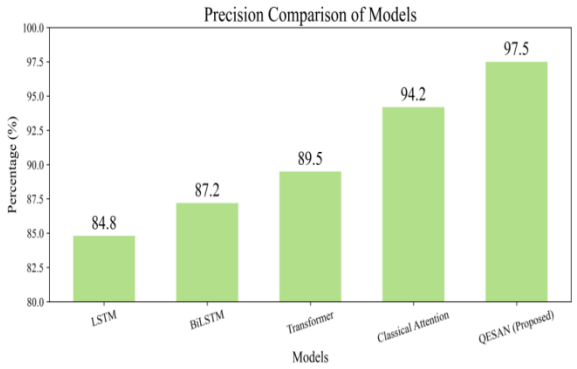


Fig 3. Precision Comparison Chart

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This figure 3 demonstrates the precision comparison of various models. The conventional models such as LSTM (84.8%), BiLSTM (87.2%), and Transformer (89.5%), and Classical Attention (94.2%), are lower and enhance the results, respectively. The proposed QESAN model is most precise (97.5%), which is the best among the rest.

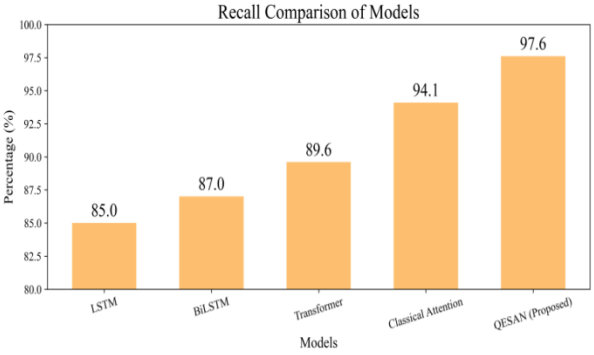


Fig 4. Recall Comparison Chat

This figure 4 makes a comparison between the memories of different models. The LSTM (85%) and BiLSTM (87%) recall, but Transformer (89.6) and Classical Attention (94.1) are more successful. QESAN proposed model has the highest recall rate (97.6 percent), and it is more effective.

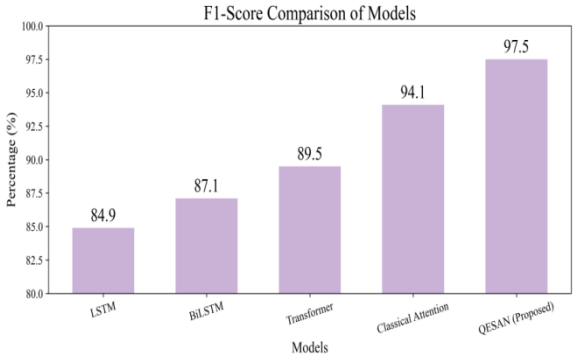


Fig 5.F1-Score Comparison Chat

This figure 5 presents the F1-score of models. The results of LSTM (84.9%) and BiLSTM (87.1%) are worse whereas the Transformer (89.5%) and Classical Attention (94.1) achieve greater results. The top F1-score (97.5%), which is what the proposed QESAN model achieves, proves that it can perform better in balancing.

V.Conclusion

The provided paper used an effective approach to resolving the issue of text classification as the seamless blend of quantum computing and artificial intelligence in the form of a Quantum-Enhanced Semantic Attention Network (QESAN). The capability to model these long-range dependencies, semantic ambiguity, or sparse or underrepresented textual information is some of such inherent issues of natural language processing which is often problematic with more existing models. The model is also an excellent model in which the complex relationships between features are captured through the process of amplitude encoding quantum superposition and transformation through entanglement, as well as semantic attention process, ensures that the greatest information in the setting is prioritized and, in such a manner, enhances its understandability and accuracy. The experiments reveal that QESAN model is immensely and remarkably better than any other model like LSTM, BiLSTM, Transformer, classical attention networks etc and it is far more accurate, precise, recalls and F1-scores. The above results imply that the proposed quantum-classical hybridization model is more predictive and scalable, predictive and has the ability to predict large and complicated textual data. Overall, the QESAN is a powerful and effective method of implementing different text classification problems because quantum computing and NLP modeling can have a core influence on the effectiveness and accuracy of models.

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